

CSE 1201 Assignment

COs	Description	Taxonomy domain/level	POs	WK	WP	EA
CO2	Apply decision-making and iterative statements in C programming to solve well-defined problems.	Cognitive/ Apply	2	4		

Problem 1: Student Result Evaluation

You are given data for N students of Varendra University. Each student has a unique Student ID and is enrolled in multiple subjects.

For each student:

- You are given the number of subjects M
- You are given marks of M subjects

Task:

For each student, compute and print:

1. Average Marks
2. Grade
 - $80 \leq \text{avg} \rightarrow \text{A}$
 - $70 \leq \text{avg} < 80 \rightarrow \text{B}$
 - $60 \leq \text{avg} < 70 \rightarrow \text{C}$
 - $50 \leq \text{avg} < 60 \rightarrow \text{D}$
 - $\text{avg} < 50 \rightarrow \text{F}$
3. Final Status
 - If any subject mark $< 40 \rightarrow \text{FAIL}$
 - Otherwise $\rightarrow \text{PASS}$

Constraints

- $1 \leq N \leq 1000, 1 \leq M_i \leq 20, 0 \leq \text{marks} \leq 100$

Input Format

```
3
VU101 5
80 75 90 85 70
VU102 4
60 55 30 65
VU103 3
45 50 55
```

Output Format

```
VU101 80.00 A PASS
VU102 52.50 D FAIL
VU103 50.00 D PASS
```

Problem 2: Smart Grocery Billing System

A supermarket wants to automate its billing system for faster checkout. Each customer buys multiple items, and each item has a price and quantity.

Your task is to compute the final bill for each customer and determine whether they qualify for a discount based on total spending.

Input Format

- The first line contains an integer T , the number of customers.

For each customer:

- The first line contains an integer N , the number of different items purchased.
- The next N lines contain:
price quantity

Task

For each customer:

1. Compute total bill:

$$\text{total} = \sum (\text{price} \times \text{quantity})$$

2. Apply discount rule:

- If $\text{total} \geq 5000 \rightarrow$ apply 10% discount
- If $\text{total} \geq 2000$ and $< 5000 \rightarrow$ apply 5% discount
- Otherwise $\rightarrow$ no discount

3. Print final payable amount (after discount).

Constraints

- $1 \leq T, N \leq 1000$
- $1 \leq \text{price}, \text{quantity} \leq 10^4$

Sample Input

```
3
2
100 10
200 5
3
500 2
300 4
100 1
1
1000 1
```

Sample Output

```
1900
2185
1000
```